

10.3.1 Writing the wave functions in real space, bosonic symmetry yields the unnormalized wave function

$$\langle \vec{r}_1, \vec{r}_2 | \Psi \rangle = \psi(\vec{r}_1)\phi(\vec{r}_2) + \psi(\vec{r}_2)\phi(\vec{r}_1).$$

We normalize this using

$$N = \langle \Psi | \Psi \rangle = 2\langle \psi | \psi \rangle \langle \phi | \phi \rangle + 2|\langle \phi | \psi \rangle|^2$$

$$\text{where } \langle \phi | \psi \rangle = \int d^3r \phi^*(\vec{r}) \psi(\vec{r}) \text{ etc.}$$

and write down

$$N^{-1/2} (\psi(\vec{r}_1)\phi(\vec{r}_2) + \psi(\vec{r}_2)\phi(\vec{r}_1)).$$

10.3.2 The normalized state vector is

$$|\Psi\rangle = \frac{1}{\sqrt{3}} (|433\rangle + |343\rangle + |334\rangle).$$

10.3.3 (1) Each particle can be in one of three states yielding $3^3 = 27$ states. If there are N particles and M states the number is M^N .

(2) Let me enumerate the possibilities in a bone-headed way. Call the states a , b , and c : You can have all three bosons in the same state which can be a , b , or c yielding three states. You can have have two of them in one state and the third in a different state. There are three possibilities for the first choice and two for the second yielding 6 states. Finally, each boson is in a different state and since they are identical (assumed) this yield one state. Thus the total is $3 + 6 + 1 = 10$.

This is clearly not the way to do N identical bosons in M states. What is the answer in this case?

(3) Each (identical) fermion is in a different state and so there is only one possibility.

10.3.4 The energy of the two-particle system is

$$E_{sys} = \frac{\pi^2 \hbar^2}{mL^2} (n_1^2 + n_2^2).$$

In the first case $n_1^2 + n_2^2 = 2$ and the only integral solution is $n_1 = n_2 = 1$. So the state vector can be written as $|\Psi\rangle = |11\rangle$. Since both particles are in the same state and there are no other degrees of freedom they are bosons. We assume that the universe has three spatial dimensions even though these particles are confined to one dimension.

In the second case $n_1^2 + n_2^2 = 5$ and the possibilities are $n_1 = 1, n_2 = 2$ or vice versa. Since these are bosons (same system but repeated measurement) the allowed wave function is

$$|\Psi\rangle = \frac{1}{\sqrt{2}} (|12\rangle + |21\rangle) .$$

10.3.6 This was discussed in class. If there are n_j identical fermions of type j , m_i bosons of type i exchange of all the constituents yields

$$(-1)^{\sum_j n_j}$$

since the bosons yield a factor of $+1$ and there is a factor of -1 for each fermion. Therefore, the sign is determined by $\sum_j n_j$ which is just the total number of fermions. This yields the stated result.

You can also demonstrate this result on the basis of the spin-statistics theorem and the rules of angular momentum addition.

15.2.2 This is a completely straightforward problem using $j_{\pm} = j_{1\pm} + j_{2\pm}$, the result

$$j_- |jm\rangle = \sqrt{(j+m)(j-m+1)} |j, m-1\rangle$$

orthogonality, and the convention that all CG coefficients are real.

(a) We start from the uniquely defined state

$$|j = 3/2, m = 3/2\rangle = |m_1 = 1, m_2 = 1/2\rangle$$

and apply j_- to both sides. This yields

$$\sqrt{3} |j = 3/2, m = 1/2\rangle = |m_1 = 1, m_2 = -1/2\rangle + \sqrt{2} |m_1 = 0, m_2 = 1/2\rangle .$$

This yields the the first non-trivial set of coefficients.

$$|j = 3/2, m = 1/2\rangle = \frac{1}{\sqrt{3}} |m_1 = 1, m_2 = -1/2\rangle + \sqrt{\frac{2}{3}} |m_1 = 0, m_2 = 1/2\rangle .$$

Since $|j = 1/2, m = 1/2\rangle$ is orthogonal to this we have

$$|j = 1/2, m = 1/2\rangle = \sqrt{\frac{2}{3}} |m_1 = 1, m_2 = -1/2\rangle - \frac{1}{\sqrt{3}} |m_1 = 0, m_2 = 1/2\rangle .$$

Denoting the Clebsch-Gordan coefficients by $C_{m_1 m_2}^{jm}$ we can use the relation

$$C_{m_1 m_2}^{jm} = (-1)^{j_1 + j_2 - j} C_{-m_1, -m_2}^{j, -m}$$

and the standard sign conventions to obtain the rest. Equivalently you can start from $|j = 3/2, m = -3/2\rangle$ and use the raising operator.

The other problem is entirely analogous. We have

$$|j = 2, m = 2\rangle = |m_1 = 1, m_2 = 1\rangle.$$

Applying j_- to both sides we have

$$|j = 2, m = 1\rangle = \frac{1}{\sqrt{2}}|m_1 = 1, m_2 = 0\rangle + \frac{1}{\sqrt{2}}|m_1 = 0, m_2 = 1\rangle.$$

By orthogonality we have

$$|j = 1, m = 1\rangle = \frac{1}{\sqrt{2}}|m_1 = 1, m_2 = 0\rangle - \frac{1}{\sqrt{2}}|m_1 = 0, m_2 = 1\rangle.$$

We apply j_- to the above two equations and obtain

$$|j = 2, m = 0\rangle = \frac{1}{\sqrt{6}}|m_1 = 1, m_2 = -1\rangle + \frac{2}{\sqrt{6}}|m_1 = 0, m_2 = 0\rangle + \frac{1}{\sqrt{6}}|m_1 = -1, m_2 = 1\rangle.$$

$$|j = 1, m = 0\rangle = \frac{1}{\sqrt{2}}|m_1 = 1, m_2 = -1\rangle - \frac{1}{\sqrt{2}}|m_1 = -1, m_2 = 1\rangle.$$

$|j = 0, m = 0\rangle$ must be orthogonal to both of the above kets and since the ket is normalized and the coefficients are real (with appropriate sign convention) we obtain

$$|j = 0, m = 0\rangle = \frac{1}{\sqrt{3}}|m_1 = 1, m_2 = -1\rangle - \frac{1}{\sqrt{3}}|m_1 = 0, m_2 = 0\rangle + \frac{1}{\sqrt{3}}|m_1 = -1, m_2 = 1\rangle.$$

The rest of the coefficients can be obtained by starting from $|j = 2, m = -2\rangle$ or by using the relations given above.

The answers are in easily available tables and in the back of the book.

15.2.7 We have discussed this in class in detail. When we add two angular momenta j_1 and j_2 the state with maximum j and m_j equal $j_1 + j_2$ is given by

$$|j = j_1 + j_2, m_j = j_1 + j_2\rangle = |m_1 = j_1, m_2 = j_2\rangle \quad ;$$

the rest of the m_j states for $j = j_1 + j_2$ are obtained by applying the symmetric (under the exchange of 1 and 2) lowering operator $J_- = J_{1-} + J_{2-}$ and thence, are symmetric. In particular,

$$|j = j_1 + j_2, m_j = j_1 + j_2 - 1\rangle = \sqrt{\frac{1}{2}}|m_1 = j_1 - 1, m_2 = j_2\rangle + \sqrt{\frac{1}{2}}|m_1 = j_1, m_2 = j_2 - 1\rangle$$

is symmetric. The state $|j = j_1 + j_2 - 1, m_j = m_1 + m_2 - 1\rangle$ is chosen to be orthogonal to the state above. Thus,

$$|j = j_1 + j_2 - 1, m_j = j_1 + j_2 - 1\rangle = \sqrt{\frac{1}{2}}|m_1 = j_1 - 1, m_2 = j_2\rangle - \sqrt{\frac{1}{2}}|m_1 = j_1, m_2 = j_2 - 1\rangle$$

which is clearly antisymmetric. The states with $j = j_1 + j_2 - 1$ with lower values of m_j are obtained by using the lowering operator and their antisymmetry is maintained.